



ELA: English Language Assistant

An AI-Assisted Framework for Authentic English Comprehension Using Recursive Linguistic Annotation

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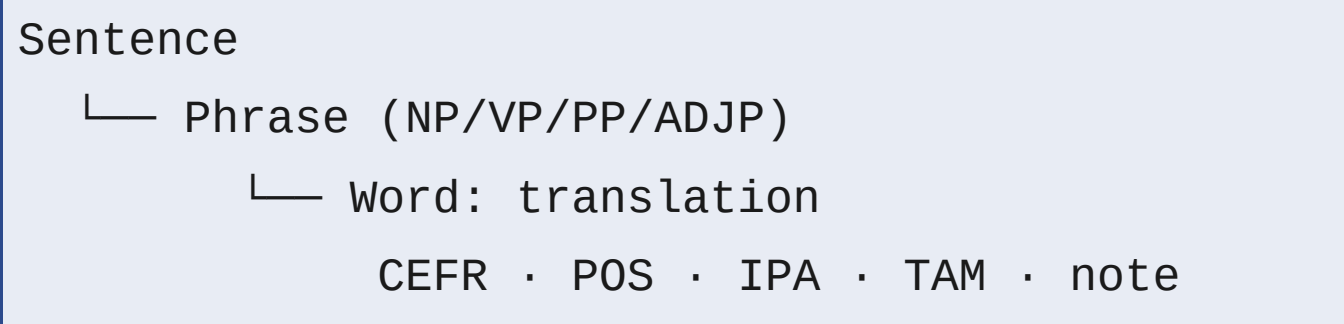
Problem & Motivation

Mainstream language learning platforms present **pre-simplified content**, stripping the grammatical structures and vocabulary range that define authentic English. Learners are shielded from complexity rather than guided through it.

ELA takes the opposite approach: authentic text and speech are annotated **in place** — **structure intact** — using rule-based NLP and fine-tuned ML models. Each sentence is decomposed into a **Recursive Linguistic Element (RLE)** tree enriched with CEFR level, tense–aspect–mood tags, IPA, and bilingual pedagogical notes.

The result is a fully offline, client-side system that bridges authentic complexity and learner comprehension — without simplification, cloud dependency, or data upload.

The RLE Data Model



Language-agnostic · EN → RU via M2M100 (offline).

Analysis Pipeline



Research Journey: What We Tried

CEFR CLASSIFICATION		NOTE GENERATION	
DeBERTa underperformed	BERT same	GPT-4o cost/cloud	T5 raw no-generalise
XGBoost ✓ 78.3%		T5+Templates ✓ ROUGE-L 0.81	
ASR		TRANSLATION	
Whisper ✓ WER 8.2%		GPT-4o opt-in	M2M100 ✓ offline

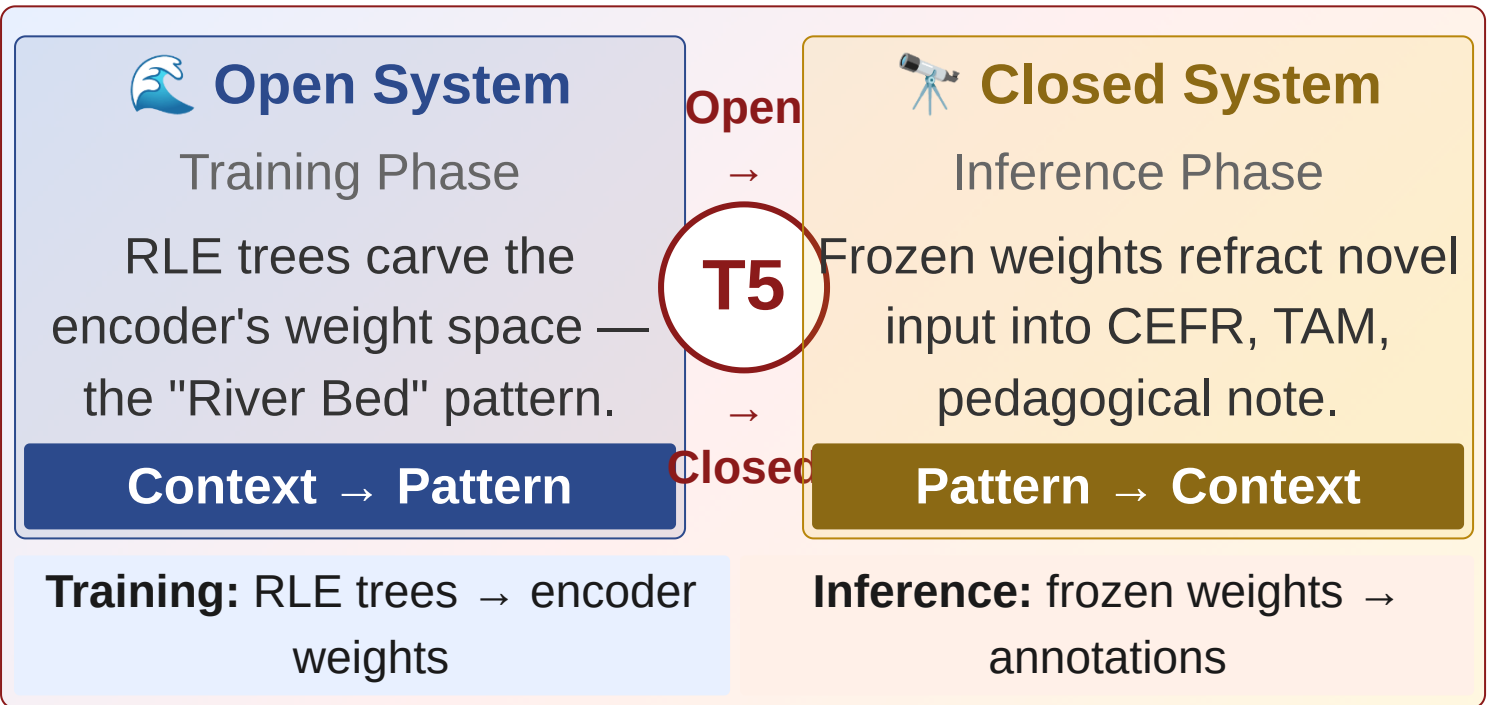
Context–Pattern Interaction

**"In open systems, context forms patterns.
In closed systems, patterns form context."**
— Discovered empirically during T5 fine-tuning

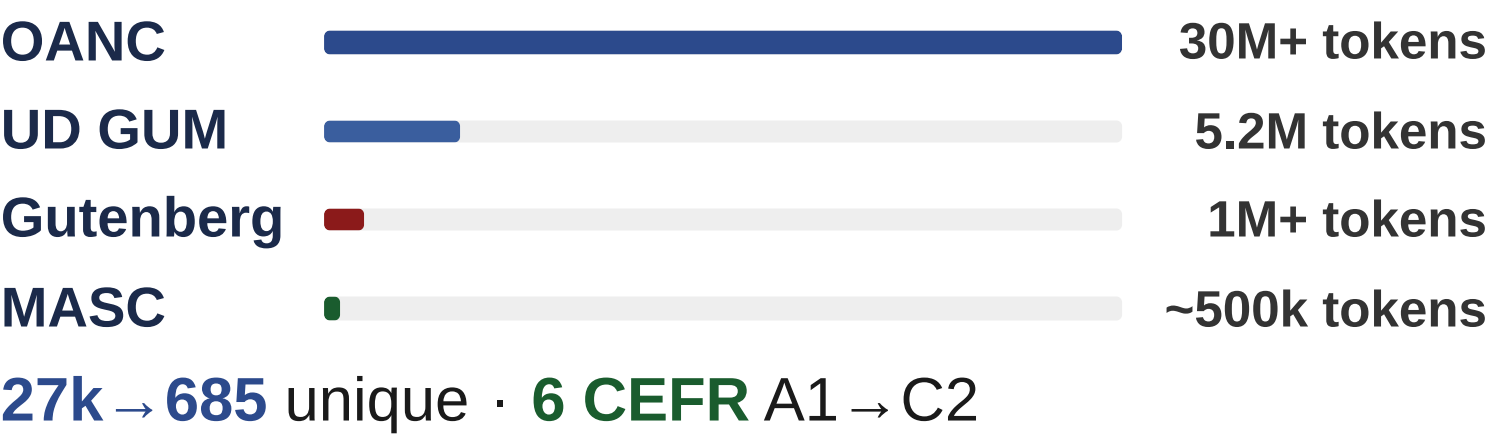
T5 on raw notation **failed**. Structural templates — **lexically invariant** — were the breakthrough:

✗ "gave him" → [VP gave][NP him]
✓ [V-PAST][IOBJ] → [TAM:past-simple]
[VOICE:active]

The Duality Loop Diagram



Training Data & Corpus Scale



Annotation Backend Comparison

Approach	Quality	Cost	Offline
GPT-4o	High	\$\$\$	✗
Rules only	Medium	Free	✓
T5 fine-tuned	High	Free	✓
XGBoost CEFR	78.3%	Free	✓

System Validation

- **O1:** spaCy + TAM → schema-valid RLE ✓
- **O2:** CEFR 78.3%; T5 ROUGE-L 0.81 ✓
- **O3:** Visualizer + JSON/CSV export ✓
- **O4:** Android PWA, fully client-side ✓

Performance Metrics

78.3% CEFR Accuracy XGBoost ensemble	0.81 ROUGE-L T5 fine-tuned	8.2% ASR WER Whisper base
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Conclusions

- AI annotation makes authentic English **accessible without simplification**.
- The **Context ↔ Pattern** duality generalises to any open → closed system; T5 placeholders are **lexically invariant**.

Future Directions

- **Tauri 2 Desktop App** — native .deb/Windows with bundled ML models.
- **Translation Quality Gates** — ChrF scorer, M2M100 review queue.
- **M2M100 LoRA Fine-Tuning** — domain-specific EN → RU refinement.
- **Real-time Streaming ASR** — Whisper streaming for live practice.

Live Demo

www.el-a.uk

Try it in your browser

1. Honnibal, M. & Montani, I. (2017). spaCy: Industrial-strength NLP.

2. Raffel, C. et al. (2020). Exploring the Limits of Transfer Learning with T5. *JMLR*.

3. Radford, A. et al. (2023). Robust Speech Recognition via Large-Scale Weak Supervision. *ICML*.

4. Chen, T. & Guestrin, C. (2016). XGBoost: A Scalable Tree Boosting System. *ACM KDD*.

5. Levy, M. (1997). *Computer-Assisted Language Learning*. Oxford University Press.

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